

PRODUCT NAME

MOTILIUM Oral Suspension 1mg/ml

QUALITATIVE AND QUANTITATIVE COMPOSITION

The oral suspension contains 1 mg domperidone per ml.

PHARMACEUTICAL FORM

Oral suspension.

CLINICAL PARTICULARS**Therapeutic Indications**

Domperidone is indicated for the relief of the symptoms of nausea and vomiting.

Nausea and vomiting of functional, organic, infectious or dietary origin.

Nausea and vomiting induced by

- radiotherapy or drug therapy.
- dopamine agonists (such as L-dopa and bromocriptine) used in the treatment of Parkinson's disease.

Dosage and Administration

It is recommended to take oral MOTILIUM 15-30 minutes before meals. If taken after meals, absorption of the drug is somewhat delayed.

1. Adults and adolescents ≥ 12 years of age and weighing ≥ 35 kg and children < 12 years of age and weighing ≥ 35 kg

The dose of MOTILIUM should be the lowest effective dose for the individual situation (typically 30 mg/day) and can be increased if necessary to a maximum daily oral dose of 40 mg.

Usually, the maximum treatment duration should not exceed one week for the treatment of acute nausea and vomiting. If nausea and vomiting persists for longer than one week, patients should consult their physician. For other indications, the initial duration of treatment is up to four weeks. If treatment exceeds four weeks, patients should be reevaluated and the need for continued treatment reassessed.

Formulation (domperidone per unit)	Dosage	Maximum dose per day
Oral suspension (1 mg/ml)	10 mL three to four times per day	40 mg (40 mL of 1 mg/mL oral suspension)

2. Adults and adolescents (≥ 12 years of age) weighing < 35 kg

The dose of MOTILIUM should be the lowest effective dose. The total daily dose is dependent on weight (see table below).

Usually, the maximum treatment duration should not exceed one week for the treatment of acute nausea and vomiting. For other indications, the initial duration of treatment is up to four

weeks. If treatment exceeds four weeks, patients should be reevaluated and the need for continued treatment reassessed. Due to the need for accurate dosing, tablets are unsuitable for use in adults and adolescents weighing less than 35kg.

Formulation (domperidone per unit)	Dosage	Maximum dose per day
Oral suspension (1 mg/mL)	0.25 mg/kg three to four times per day	35 mg (1 mg/kg but no more than 35 mL)

3. Infants and children < 12 years of age and weighing < 35 kg

The efficacy of MOTILIUM has not been established in infants and children < 12 years of age and weighing < 35 kg (see *Clinical Studies*).

Renal impairment

Since the elimination half-life of domperidone is prolonged in severe renal impairment (serum creatinine > 6 mg/100 mL, i.e. > 0.6 mmol/L), the dosing frequency of MOTILIUM should be reduced to once or twice daily, depending on the severity of the impairment, and the dose may need to be reduced. Patients with severe renal impairment should be reviewed regularly (see *Pharmacokinetic Properties*).

Hepatic impairment

MOTILIUM is contraindicated for patients with moderate (Child-Pugh 7 to 9) or severe (Child-Pugh >9) hepatic impairment (see *Contraindications*). Dose adjustment is not required for patients with mild (Child-Pugh 5 to 6) hepatic impairment (see *Pharmacokinetic Properties*).

Contraindications

MOTILIUM is contraindicated in the following situations:

- Known hypersensitivity to domperidone or any of the excipients.
- Prolactin-releasing pituitary tumour (prolactinoma).
- In patients who have known existing prolongation of cardiac conduction intervals, particularly QTc, patients with significant electrolyte disturbances or underlying cardiac diseases such as congestive heart failure (see *Warnings and Precautions*).
- co-administration with QT-prolonging drugs (see *Interactions*)
- co-administration with potent CYP3A4 inhibitors (regardless of their QT-prolonging effects) (see *Interactions*).
- Whenever stimulation of gastric motility might be dangerous, e.g., in the presence of gastro-intestinal haemorrhage, mechanical obstruction or perforation.
- In patients with moderate or severe hepatic impairment (See *Pharmacokinetic Properties*).

Warnings and Precautions

Cardiovascular effects

Domperidone has been associated with prolongation of the QT interval on the electrocardiogram. During post-marketing surveillance, there have been very rare cases of QT-prolongation and torsades de pointes in patients taking domperidone. These reports included patients with confounding risk factors, electrolyte abnormalities and concomitant treatment which may have been contributing factors (see *Adverse Reactions*).

Epidemiological studies showed that domperidone was associated with an increased risk of serious ventricular arrhythmias or sudden cardiac death (see *Adverse Reactions*). A higher risk was observed in patients older than 60 years, patients taking daily doses greater than 30 mg, and patients concurrently taking QT-prolonging drugs or CYP3A4 inhibitors.

Domperidone should be used at the lowest effective dose in adults and children.

Domperidone is contraindicated in patients with known existing prolongation of cardiac conduction intervals, particularly QTc, in patients with significant electrolyte disturbances (hypokalaemia, hyperkalaemia, hypomagnesaemia), or bradycardia, or in patients with underlying cardiac diseases such as congestive heart failure due to increased risk of ventricular arrhythmia (see *Contraindications*).

Electrolyte disturbances (hypokalaemia, hyperkalaemia, hypomagnesaemia) or bradycardia are known to be conditions increasing the proarrhythmic risk.

Treatment with domperidone should be stopped if signs or symptoms occur that may be associated with cardiac arrhythmia, and the patients should consult their physician.

Patients should be advised to promptly report any cardiac symptoms.

Drug interaction potential

The main metabolic pathway of domperidone is through CYP3A4. *In vitro* and human data show that the concomitant use of drugs that significantly inhibit this enzyme may result in increased plasma levels of domperidone. Co-administration of domperidone with potent CYP3A4 inhibitors which have been shown to cause QT interval prolongation is contraindicated (see *Contraindications*).

Caution should be exercised when domperidone is co-administered with potent CYP3A4 inhibitors which have not been shown to cause QT interval prolongation such as indinavir and patients should be monitored closely for signs or symptoms of adverse reactions (see *Adverse Reactions*).

Caution should be exercised when domperidone is co-administered with drugs which have been shown to cause QT interval prolongation and patients should be monitored closely for signs or symptoms of cardiovascular adverse reactions (see *Adverse Reactions*). Examples include:

- anti-arrhythmics class IA (e.g., disopyramide, quinidine)
- anti-arrhythmics class III (e.g., amiodarone, dofetilide, dronedarone, ibutilide, sotalol)
- certain antipsychotics (e.g., haloperidol, pimozide, sertindole)
- certain antidepressants (e.g., citalopram, escitalopram)
- certain antibiotics (e.g., levofloxacin, moxifloxacin)
- certain antifungal agents (e.g., pentamidine)
- certain antimalarial agents (e.g., halofantrine)
- certain gastro-intestinal drugs (e.g., dolasetron)
- certain drugs used in cancer (e.g., toremifene, vandetanib)
- certain other drugs (e.g., bepridil, methadone)

Antacids or antisecretory agents should not be taken simultaneously with oral formulations of MOTILIUM, as they lower the oral bioavailability of domperidone. When used concomitantly, MOTILIUM should be taken before meals and antacids or antisecretory agents after meals.

Excipients

MOTILIUM oral suspension contains sorbitol and may be unsuitable for patients with sorbitol intolerance.

Interactions

The main metabolic pathway of domperidone is through CYP3A4. *In vitro* and human data show that the concomitant use of drugs that significantly inhibit this enzyme may result in increased plasma levels of domperidone.

When domperidone was co-administered with potent CYP3A4 inhibitors which have been shown to cause QT interval prolongation, clinically relevant changes in QT intervals were observed. Therefore, co-administration of domperidone with certain drugs is contraindicated (see Contraindications).

Concomitant administration of anticholinergic drugs (e.g., dextromethorphan, diphenhydramine) may antagonize the anti-dyspeptic effect of MOTILIUM.

Theoretically, since MOTILIUM has gastro-kinetic effects it could influence the absorption of concomitantly orally administered drugs, particularly those with sustained release or enteric coated formulations. However, in patients already stabilised on digoxin or paracetamol, concomitant administration of domperidone did not influence the blood levels of these drugs.

MOTILIUM may also be given with:

- neuroleptics, the action of which it does not potentiate,
- dopaminergic agonists (bromocriptine, L-dopa), whose unwanted peripheral effects such as digestive disorders, nausea and vomiting it suppresses without counteracting their central properties.

Pregnancy and Breast-feeding

Pregnancy

There are limited post-marketing data on the use of domperidone in pregnant women. A study in rats has shown reproductive toxicity at a high, maternally toxic dose. The potential risk for humans is unknown. Therefore, MOTILIUM should only be used during pregnancy when justified by the anticipated therapeutic benefit.

Breast-feeding

The amount of domperidone that could be ingested by an infant through breast milk is low. The maximal relative infant dose (%) is estimated to be about 0.1% of the maternal weight- adjusted dosage. It is not known whether this is harmful to the newborn. Therefore, breast- feeding is not recommended for mothers who are taking MOTILIUM.

Effects on Ability to Drive and Use Machines

Dizziness and somnolence have been observed following use of domperidone (see Adverse Reactions). Therefore, patients should be advised not to drive or use machinery or engage in other activities requiring mental alertness and coordination until they have established how MOTILIUM affects them.

Adverse Reactions

Throughout this section, adverse reactions are presented. Adverse reactions are adverse events that were considered to be reasonably associated with the use of domperidone based on the comprehensive assessment of the available adverse event information. A causal relationship with domperidone cannot be reliably established in individual cases. Further, because clinical trials are conducted under widely varying conditions, adverse reaction rates

observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice.

Clinical Trial Data

The safety of MOTILIUM was evaluated in 1221 patients with gastroparesis, dyspepsia, gastro-oesophageal reflux disorder (GERD), or other related conditions in 45 clinical trials included in the safety database. All patients were ≥ 15 years old and received at least one dose of oral domperidone base. Slightly fewer than one-half (553/1221) of patients were diabetic. The median total daily dose was 80 mg (range 10 to 160 mg), with 230 patients receiving a dose greater than 80 mg. Median duration of exposure was 56 days (range 1 to 2248 days).

ARs reported by $\geq 1\%$ of patients treated with domperidone in these 45 clinical trials are shown in Table 1.

Table 1. Adverse Reactions Reported by $\geq 1\%$ of Domperidone -Treated Patients in 45 Clinical Trials	
System/Organ Class	Domperidone
Adverse Reaction	(n=1221)
	%
Psychiatric Disorders	
Depression	2.5
Anxiety	1.6
Libido Decreased/Loss of Libido	1.5
Nervous System Disorders	
Headache	5.6
Somnolence	2.5
Akathisia	1.0
Gastrointestinal Disorders	
Diarrhoea	5.2
Skin and Subcutaneous Tissue Disorders	
Rash	2.8
Pruritus	1.7
Reproductive System and Breast Disorders	
Breast Enlargement/Gynaecomastia	5.3
Breast Tenderness	4.4
Galactorrhoea	3.3
Amenorrhoea	2.9
Breast Pain	2.3
Menstruation Irregular	2.0
Lactation Disorder	1.6
General Disorders and Administration Site Conditions	
Asthenia	1.9

ARs that occurred in $<1\%$ of domperidone-treated patients in the 45 clinical trials (n=1221) are listed below in Table 2.

Table 2. Adverse Reactions Reported by <1% Domperidone-Treated Patients in 45 Clinical Trials

System/Organ Class Adverse Reaction	Domperidone (n=1221) %
Immune System Disorders	
Hypersensitivity	0.2
Skin and Subcutaneous Tissue Disorders	
Urticaria	0.7
Reproductive System and Breast Disorders	
Breast Discharge	0.8
Breast Swelling	0.5

The following adverse reaction has been reported with over-the-counter use: dry mouth.

Postmarketing

In addition to the ARs reported during clinical studies and listed above, the following ARs have been reported during postmarketing experience (Table 3). In the table, the frequencies are provided according to the following convention:

Very common	≥ 1/10
Common	≥ 1/100 and < 1/10
Uncommon	≥ 1/ 1,000 and < 1/100
Rare	≥1/10,000 and < 1/1,000
Very rare	<1/10,000, including isolated reports.

In Table 3, ARs are presented by frequency category based on spontaneous reporting rates.

Table 3. Adverse Reactions Identified During Postmarketing Experience with Domperidone by Frequency Category Estimated from Spontaneous Reporting Rates

Immune System Disorders	
<i>Very rare</i>	Anaphylactic Reaction (including Anaphylactic Shock)
Psychiatric Disorders	
<i>Very rare</i>	Agitation, Nervousness
Nervous System Disorders	
<i>Very rare</i>	Dizziness, Extrapyrarnidal Disorder, Convulsion
Cardiac Disorders	
<i>Very rare</i>	Ventricular arrhythmias, QTc prolongation, Torsade de Pointes, Sudden cardiac death (see Warnings and Precautions)
Skin and Subcutaneous Tissue Disorders	
<i>Very rare</i>	Angioedema
Renal and Urinary Disorders	
<i>Very rare</i>	Urinary Retention
Investigations	
<i>Very rare</i>	Liver Function Test Abnormal, Blood Prolactin Increased

*Based on epidemiology data

Pediatric population

In postmarketing experience, there were no differences in the safety profile of adults and children.

Overdose

Symptoms and signs

Overdose has been reported primarily in infants and children. Symptoms of overdose may include agitation, altered consciousness, convulsion, disorientation, somnolence and extrapyramidal reactions.

Treatment

There is no specific antidote to domperidone. Close medical supervision and supportive therapy is recommended. Anticholinergic or anti-Parkinson drugs may be helpful in controlling the extrapyramidal reactions.

It is advisable to contact a poison control center to obtain the latest recommendations for the management of an overdose.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic Properties

Pharmacotherapeutic group: Propulsive, ATC code: A03FA03

Domperidone is a dopamine antagonist with anti-emetic properties. Domperidone does not readily cross the blood-brain barrier. In domperidone users, especially in adults, extrapyramidal side effects are very rare, but domperidone promotes the release of prolactin from the pituitary. Its anti-emetic effect may be due to a combination of peripheral (gastrokinetic) effects and antagonism of dopamine receptors in the chemoreceptor trigger zone, which lies outside the blood-brain barrier in the area postrema. Animal studies, together with the low concentrations found in the brain, indicate a predominantly peripheral effect of domperidone on dopamine receptors.

Studies in man have shown oral domperidone to increase lower esophageal pressure, improve antroduodenal motility and accelerate gastric emptying. There is no effect on gastric secretion.

Effect on QT/QTc Interval and Cardiac Electrophysiology

In accordance with ICH—E14 guidelines, a thorough QT study was performed in healthy subjects. This study included a placebo, active comparator and positive control and was conducted using recommended and supra-therapeutic doses (10 and 20 mg administered 4 times a day). This study found a maximal difference of QTc between domperidone and placebo in LS-means in the change from baseline of 3.4 msec for 20 mg domperidone administered 4 times a day on Day 4, and the 2-sided 90% CI (1.0 5.9 msec) did not exceed 10 msec. The QT prolongation observed in this study when domperidone was administered according to the recommended dosing regimen is not clinically relevant.

This lack of clinical relevance is corroborated by pharmacokinetics and QTc interval data from two older studies which involved a 5-day treatment of 20 mg and 40 mg domperidone administered 4 times a day. ECGs were recorded prior to the study, on Day 5 at 1 hour (approximately at t_{max}) after the morning dose, and 3 days later. In both studies, no difference between QTc after active treatment and placebo was observed. It was therefore concluded that domperidone administration of 80 and 160 mg daily doses had no clinically significant effect on QTc in healthy subjects.

Clinical Studies

Infants and children ≤ 12 years of age

A multicenter, double-blind, randomized, placebo-controlled, parallel-group, prospective study was conducted to evaluate the safety and efficacy of domperidone in 292 children with acute gastroenteritis aged 6 months to 12 years (median age 7 years). In addition to oral rehydration treatment (ORT), randomized subjects received domperidone oral suspension at 0.25 mg/kg (up to a maximum of 30 mg domperidone/day), or placebo, 3 times a day, for up to 7 days. This study did not achieve the primary objective, which was to demonstrate that domperidone suspension plus ORT is more effective than placebo plus ORT at reducing the percentage of subjects with no vomiting episodes during the first 48 hours after the first treatment administration.

Pharmacokinetic Properties

Absorption

In fasting subjects, domperidone is rapidly absorbed after oral administration, with peak plasma concentrations occurring at approximately 60 minutes after dosing. The key pharmacokinetic parameters after a single or multiple doses (administered 4 times a day) of 10 mg domperidone base tablets to healthy subjects are presented in the table below. The C_{max} and AUC values of domperidone increased proportionally with dose in the 10 mg to 20 mg dose range.

Key Domperidone Pharmacokinetic Parameters After a Single or Multiple Doses (administered 4 times a day) of 10 mg Domperidone Base Tablets to Healthy Subjects		
PK parameter	Domperidone 10 mg administered four times a day	
Mean	Day 1	Day 4
n	40	40
C _{min} , ng/mL	NA	5.26 (CV: 31.1%)
C _{max} , ng/mL	11.6 (CV: 50.8%)	17.3 (CV: 35.4%)
t _{max} , h ^a	1.02 (range: 0.52 - 5.02)	1.02 (range: 0.50 - 4.03)
AUC _{5h} , ng.h/mL	20.4 (CV: 34.4%)	47.8 (CV: 30.5%)
^a median (range)		
Source: Study DOM-DYP-1001		

The low absolute bioavailability of oral domperidone (approximately 15%) is due to an extensive first-pass metabolism in the gut wall and liver. Although domperidone's bioavailability is enhanced in normal subjects when taken after a meal, patients with gastrointestinal complaints should take domperidone 15 - 30 minutes before a meal. Reduced gastric acidity impairs the absorption of domperidone base. Oral bioavailability of domperidone base is decreased by prior concomitant administration of cimetidine and sodium bicarbonate. The time of peak absorption is slightly delayed and the AUC somewhat increased when the oral drug is taken after a meal.

Distribution

Domperidone is 91 - 93% bound to plasma proteins. Distribution studies with radiolabelled drug in animals have shown wide tissue distribution, but low brain concentration. Small amounts of drug cross the placenta in rats.

Metabolism

Domperidone undergoes rapid and extensive hepatic metabolism by hydroxylation and N-dealkylation. *In vitro* metabolism experiments with diagnostic inhibitors revealed that CYP3A4 is a major form of cytochrome P-450 involved in the N-dealkylation of domperidone, whereas CYP3A4, CYP1A2 and CYP2E1 are involved in domperidone aromatic hydroxylation.

Excretion

Urinary and faecal excretions amount to 31 and 66% of the oral dose respectively. The proportion of the drug excreted unchanged is small (10% of faecal excretion and approximately 1% of urinary excretion).

The plasma half-life after a single oral dose is 7 - 9 hours in healthy subjects, but is prolonged in patients with severe renal insufficiency.

Special Populations

Hepatic Impairment

In subjects with moderate hepatic impairment (Pugh score 7 to 9, Child-Pugh rating B), the AUC and C_{max} of domperidone is 2.9- and 1.5-fold higher, respectively, than in healthy subjects. The unbound fraction is increased by 25%, and the terminal elimination half-life is prolonged from 15 to 23 hours. Subjects with mild hepatic impairment have a somewhat lower systemic exposure than healthy subjects based on C_{max} and AUC, with no change in protein binding or terminal half-life. Subjects with severe hepatic impairment were not studied. (See Contraindications.)

Renal Impairment

In subjects with severe renal insufficiency (serum creatinine > 6 mg/100 mL, i.e., > 0.6 mmol/L) the half-life of domperidone is increased from 7.4 to 20.8 hours, but plasma drug levels are lower than in subjects with normal renal function. Very little unchanged drug (approximately 1%) is excreted via the kidneys. (See Dosage and Administration.)

Non-Clinical Information

At a high, maternally toxic dose of 200 mg/kg/day, teratogenic effects (organ abnormalities such as anophthalmia, microphthalmia and displacement of the subclavian artery) were seen in the rat. The clinical significance of these findings is unknown. No teratogenicity was observed in mice and rabbits.

Electrophysiological *in vitro* and *in vivo* studies have shown that domperidone, at high concentrations, may prolong the QTc interval.

In juvenile rats, a no observed adverse effect level of 10 mg/kg was observed following 30 days of once daily repeat intraperitoneal dosing. Single intraperitoneal or intravenous doses showed similar LD₅₀ values (mean range 53-76 mg/kg) in both juvenile and adult rats.

PHARMACEUTICAL PARTICULARS

List of Excipients

Oral Suspension

Sodium saccharin, sodium carboxy-methylcellulose, sorbitol, methylparahydroxybenzoate, propylparahydroxy-benzoate, sodium hydroxide, polysorbate 20, purified water.

Incompatibilities

None Known.

Shelf-life

3 years.

Special Precautions for Storage

Store below 30°C.

Keep out of sight and reach of children.

Nature and Contents of Container

100 and 200 ml glass bottles with a measuring cap of 10 ml or a sucking pipette 5 ml.

Instructions for Use and Handling

Mix the contents of the bottle completely using a gentle tilting motion to avoid the formation of foam.

Directions for opening the bottle:

Fig. 1: The bottle comes with a childproof cap, and should be opened as follows:

- Push the plastic screw cap down while turning it counter clockwise.
- Remove the unscrewed cap.

Directions for using the pipette (supplied with the 100 ml bottle only):

Fig. 2: Insert the pipette into the bottle.

While holding the bottom ring, pull the top ring up to the mark corresponding to your weight in kilograms.

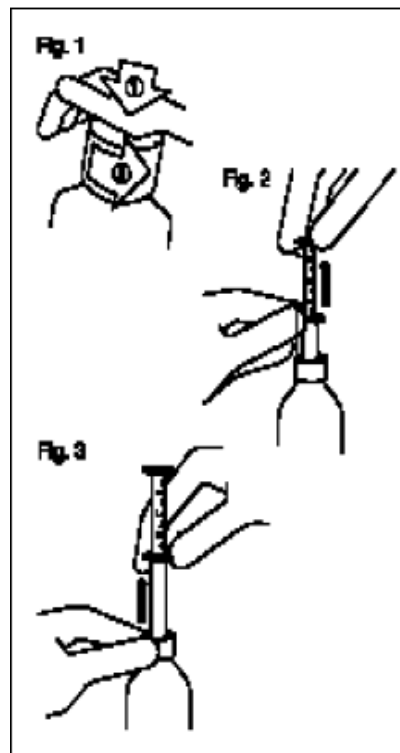
Fig. 3: Holding the bottom ring, remove the entire pipette from the bottle.

Empty the pipette.
Close the bottle.

Rinse the pipette with some water.

Directions for using the measuring cup

A measuring cup is supplied with the MOTILIUM oral solution. Use the measuring cup just as it sits on the bottle. Make sure that the side with the graduations (the side that holds less) is uppermost; that is the side you have to fill. When the arrow on the side points up, the correct side is uppermost.



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